

Project Summary

A leading logistics company is currently facing challenges in maintaining delivery efficiency, ensuring customer satisfaction, and optimizing shipment performance across different transport modes and regions. Despite growing demand, the organization has observed inconsistencies in shipment completion rates, delays in high-priority deliveries, and varied levels of customer feedback, particularly tied to certain modes of transport such as Motorbikes and Lorries. Regional disparities in delivery times and shipment value have also raised concerns about route planning and carrier performance.

As a data analyst, you have been brought in to perform a comprehensive analysis of the company's logistics data spanning from 2020 to 2024. Your task is to uncover insights that will help decision-makers understand the underlying causes of these issues and support data-driven improvements to the logistics operation.

Structure your analysis around three key focus areas:

1. Track shipment volumes, identify trends in completed versus in-progress shipments, and assess delivery performance across time.
2. Break down expected shipment volumes by transport type (Jet, Bus, Lorry, Motorbike), priority level, and geographic distribution to reveal demand patterns and bottlenecks.
3. Evaluate customer satisfaction scores, payment status, shipment value, and delivery time to determine their influence on service quality perceptions.

Your objective is to deliver a clear, actionable assessment that highlights inefficiencies, identifies opportunities for route or process optimization, and supports executive decisions on resource allocation, carrier management, and customer experience strategies. Use this analysis to recommend improvements that will enhance delivery timelines, reduce operational costs, and improve overall satisfaction across the logistics network.

Goal

The goal of this project is to deliver a holistic view of the company's logistics performance by leveraging key performance indicators such as delivery time, shipment value, and shipment status. The analysis will aim to reveal underlying trends and performance fluctuations across different transport modes and regions, helping to pinpoint areas of strength and those requiring improvement. Ultimately, the project

seeks to provide actionable insights that will enable stakeholders to make informed decisions, enhance operational efficiency, streamline delivery processes, and improve overall customer satisfaction.

Project Objectives

- 1. Quantify completed and in-progress shipments across multiple transport modes (Jet, Bus, Lorry, Motorbike).
- 2. Evaluate shipment value trends and the financial performance of logistics services.
- 3. Analyze shipment efficiency and delivery performance across key regions (Lagos, Abuja, Port Harcourt, etc.).
- 4. Measure customer satisfaction levels, identify drivers of positive or negative experiences, and examine the impact of shipment delays or payment status on ratings.
- 5. Recommend improvements to enhance shipment delivery timelines, optimize transport mode deployment, and increase overall satisfaction.

Data Dictionary

Column Name	Description
Shipment ID	Unique identifier for each shipment.
Shipment Status	Current status of the shipment (e.g., Completed, Ready for Pickup, In Transit, Not Completed).
Order Date	The date when the shipment was ordered.
Delivery Date	The actual delivery date of the shipment.
Shipping Mode	Mode of transport used (Jets, Bus, Lorry, Motorbike).
Carrier Name	Name of the logistics carrier handling the shipment.
Shipment Weight (lbs)	Weight of the shipment in pounds.
Delivery Time (Days)	Number of days it took from order to delivery.
Delivery Location	The final destination of the shipment (city and country).
Shipment Cost/Weight (\$)	Cost per unit weight of the shipment.
Loading Time (hrs)	Time taken to load the shipment, in hours.
Loading Weight (lbs)	Weight of shipment at loading, may vary slightly from shipment weight.
Estimated Delivery Date	Forecasted delivery date at the time of order.
Tracking Number	Unique tracking number assigned to the shipment.
Shipment Type	Classification such as Standard, Express, or Urgent.
Customer Type	Type of customer (Business, Retail, International).
Payment Status	Indicates whether the payment is Paid, Pending, or Overdue.

Shipment Value (\$)	Monetary value of the shipment.
Shipment Priority	Priority level (High, Medium, Low).
Carrier Performance Rating	Carrier service rating (scale of 1–5).
Delivery Window	Preferred delivery window (e.g., 9 AM – 12 PM).
Customer Satisfaction Score	Satisfaction rating (scale of 1–10).

Deliverables

- 1. Power BI Dashboard or Presentation Report
- 2. Summary Insights & Recommendations Document
- 3. Executive Summary Slide Deck